

Choose the basis for the physics you need

- **Valence flexibility:** use double-, triple-, or higher- ζ descriptions to give each valence orbital several radial shapes
- **Polarization functions:** add higher angular momenta to describe anisotropic density, bonding, and angular correlation
- **Diffuse functions:** add small exponents to describe long-range density
 - anions, Rydberg states, noncovalent interactions, and response properties
- **Tight functions:** add large exponents to describe the core region
 - core correlation and properties sensitive to the nucleus

Practical rule

The appropriate basis depends on both the **electronic-structure method** and the **property of interest**.

Basis Set Exchange: the practical starting point

<https://www.basissetexchange.org>

What can you find there?

Basis-set definitions, element coverage, references, and input formats for most electronic-structure programs.

STO- n G: approximate Slater orbitals with Gaussians

$$\chi_{\mu}(\mathbf{r}) = \sum_{i=1}^n d_{\mu i} g_{\mu i}(\mathbf{r}) \quad (1)$$

- **Minimal basis:** one contracted basis function for each AO in the minimal atomic description
- Each contracted function contains n primitive GTOs
- The contraction coefficients and exponents are chosen so that the contraction approximates a Slater-type orbital
- STO- n G basis sets are therefore **minimal, single- ζ basis sets**

Example

STO-3G uses three primitive Gaussians to approximate each minimal-basis atomic orbital.

Pople names are construction recipes

$$6-31+G^*$$

- 6 Six primitive GTOs contracted into each core shell
- 31 Split-valence double- ζ description: one contraction of three primitives and one uncontracted primitive
- + One set of diffuse functions on non-hydrogen atoms
- * One set of polarization functions on non-hydrogen atoms, $* \equiv (d)$

Pople suffixes add flexibility

More radial flexibility

6-311++G**

- 311: split-valence triple- ζ description
- ++: diffuse functions on both heavy atoms and H
- **: polarization on both heavy atoms and H, $** \equiv (d, p)$

Explicit polarization specification

6-311G(2df,2pd)

- 2d1f on non-hydrogen atoms
- 2p1d on hydrogen atoms

Dunning basis sets systematically approach the CBS limit

Correlation-consistent hierarchy

cc-pVXZ

$X = \text{D, T, Q, 5, 6, } \dots$

- Designed primarily for correlated wave-function methods
- Polarization functions are added in balanced groups
- Increasing X systematically improves the valence space

Diffuse density

aug-, d-aug-, and t-aug-cc-pVXZ

Core correlation

cc-pCVXZ and cc-pwCVXZ

$$E_c(X) \approx E_c^{\text{CBS}} + AX^{-3} \quad (2)$$

A commonly used two-point extrapolation model

Cardinal number and augmentation add complementary flexibility

Contracted shells for second-period p-block atoms (B–Ne)

	DZ	TZ	QZ
cc-pVXZ	3s2p1d	4s3p2d1f	5s4p3d2f1g
aug-cc-pVXZ	3s2p1d+1s1p1d	4s3p2d1f+1s1p1d1f	5s4p3d2f1g+1s1p1d1f1g
d-aug-cc-pVXZ	3s2p1d+2s2p2d	4s3p2d1f+2s2p2d2f	5s4p3d2f1g+2s2p2d2f2g

- Valence and polarization space
- One diffuse layer
- Two diffuse layers

Important

The notation counts *radial shells*, not individual basis functions: one spherical *d* shell contains five functions.

Core–valence sets add tight correlation functions

Contracted shells for second-period p-block atoms (B–Ne)

	Valence set	Tight functions added	Core–valence set
DZ	3s2p1d	1s1p	4s3p1d
TZ	4s3p2d1f	2s2p1d	6s5p3d1f
QZ	5s4p3d2f1g	3s3p2d1f	8s7p5d3f1g

cc-pCVXZ

Optimized to recover both **core–core** and **core–valence** correlation.

cc-pwCVXZ

Uses a weighted optimization that emphasizes **core–valence** correlation.

Frozen-core calculations

If the core orbitals are not correlated, the additional tight core-correlation functions are usually unnecessary.

Jensen basis sets target the mean-field basis-set limit

Carbon example

Basis	Contracted shells
pc-0	3s2p
pc-1	3s2p1d
pc-2	4s3p2d1f
pc-3	6s5p4d2f1g

Polarization-consistent hierarchy

- Designed for efficient convergence of HF and KS-DFT calculations
- Higher angular momenta are selected according to their energetic importance
- Diffuse variants: aug-pc- n
- Segmented variants: pcseg- n

$$E_{\text{HF}}(n) \approx E_{\text{HF}}^{\text{CBS}} + A(n+1) \exp(-B\sqrt{n}) \quad (3)$$

One extrapolation model proposed for the pc- n hierarchy